In [4]:
```
import pandas as pd

# Load dataset
df = pd.read_csv('data_pertanian.csv')

# Tampilkan 10 data pertama
df.head(10)
```

Out[4]:

	suhu	kelembaban	curah_hujan	luas_lahan	hasil_panen	kategori_hasil
0	25.618102	61.100021	192.609494	0.732568	43.242477	sedang
1	34.260715	82.274364	25.241989	2.891096	35.081486	rendah
2	30.979909	71.002459	48.488614	2.932858	35.572948	rendah
3	28.979877	77.799974	269.566257	3.368435	57.326919	tinggi
4	22.340280	91.764827	181.928718	3.767411	51.431076	tinggi
5	22.339918	68.725228	2.759115	4.891334	30.245315	rendah
6	20.871254	74.363402	30.441463	2.823352	30.018898	rendah
7	32.992642	86.444290	199.050531	1.953304	52.188626	tinggi
8	29.016725	68.007936	1.518475	4.078338	28.978687	rendah
9	30.621089	62.694297	48.242415	1.718745	34.171692	rendah

In [2]:
```
import pandas as pd

pd.set_option('display.max_rows', None)
pd.set_option('display.max_columns', None)

df
```

Out[2]:

	suhu	kelembaban	curah_hujan	luas_lahan	hasil_panen	kategori_hasil
0	25.618102	61.100021	192.609494	0.732568	43.242477	sedang
1	34.260715	82.274364	25.241989	2.891096	35.081486	rendah
2	30.979909	71.002459	48.488614	2.932858	35.572948	rendah
3	28.979877	77.799974	269.566257	3.368435	57.326919	tinggi
4	22.340280	91.764827	181.928718	3.767411	51.431076	tinggi
5	22.339918	68.725228	2.759115	4.891334	30.245315	rendah
6	20.871254	74.363402	30.441463	2.823352	30.018898	rendah
7	32.992642	86.444290	199.050531	1.953304	52.188626	tinggi
8	29.016725	68.007936	1.518475	4.078338	28.978687	rendah
9	30.621089	62.694297	48.242415	1.718745	34.171692	rendah
10	20.308767	70.141301	164.620137	2.475371	39.521612	rendah
11	34.548648	65.642745	207.568559	0.853054	43.527729	sedang
12	32.486640	92.539418	195.588378	0.614078	51.357092	tinggi
13	23.185087	88.284213	67.280793	4.831918	42.587609	sedang
14	22.727375	82.169131	213.653766	4.261911	54.389476	tinggi
15	22.751068	90.501121	71.174726	3.631884	40.563476	sedang
16	24.563634	88.128523	97.619909	2.340288	39.412869	rendah
17	27.871346	66.529952	223.947422	1.279824	44.827277	sedang
18	26.479175	91.239565	194.889870	1.203967	48.240195	sedang
19	24.368437	78.876978	254.767023	1.626093	50.460492	tinggi
20	29.177793	88.260405	197.283868	2.971520	54.027085	tinggi
21	22.092408	91.363195	170.492581	3.715682	49.086868	sedang
22	24.382170	71.130122	28.102430	3.470888	29.641700	rendah
23	25.495428	63.851817	110.314741	1.759703	34.327099	rendah
24	26.841050	67.977731	79.560710	4.796894	40.023583	rendah
25	31.777639	74.948773	73.196893	3.820536	38.356359	rendah
26	22.995107	88.630517	291.903166	2.994593	58.159698	tinggi
27	27.713517	90.125570	117.929317	3.252743	45.124962	sedang
28	28.886219	60.243325	267.613967	2.388200	52.742261	tinggi
29	20.696756	77.876156	189.341588	1.614789	42.934109	sedang

	suhu	kelembaban	curah_hujan	luas_lahan	hasil_panen	kategori_hasil
30	29.113173	74.609385	238.443391	2.101877	50.761845	tinggi
31	22.557862	67.773773	150.791128	3.910307	43.685941	sedang
32	20.975774	64.195288	173.071165	0.564771	34.672279	rendah
33	34.233283	71.816531	147.755308	1.022327	38.638548	rendah
34	34.484480	93.001840	58.572896	0.707012	34.780137	rendah
35	32.125960	71.312103	216.735635	0.683280	46.513437	sedang
36	24.569207	78.157672	84.231709	4.349573	40.746428	sedang
37	21.465082	84.605664	7.294790	3.666460	34.373769	rendah
38	30.263495	72.727036	193.641689	2.633782	49.971508	tinggi
39	26.602287	94.012373	53.133204	0.940254	33.657112	rendah
40	21.830574	93.685655	282.137575	2.712271	58.886571	tinggi
41	27.427654	68.812380	286.178573	2.630623	53.864817	tinggi
42	20.515828	77.403698	274.459317	1.279408	51.603210	tinggi
43	33.639806	70.530741	111.047610	2.452332	39.630199	rendah
44	23.881700	69.969417	4.636985	2.293271	26.854072	rendah
45	29.937834	61.291043	278.495569	3.271325	53.977305	tinggi
46	24.675666	81.334752	128.455244	3.357921	44.269711	sedang
47	27.801020	77.593766	289.996446	0.703868	57.331918	tinggi
48	28.200654	61.801756	289.085993	2.185757	53.883140	tinggi
49	22.772817	69.752626	255.902837	3.316370	53.808817	tinggi
50	34.543769	91.789306	88.334668	2.764113	44.462973	sedang
51	31.626992	68.384666	115.529319	4.354204	42.623930	sedang
52	34.092484	65.071321	255.341001	3.464121	56.152537	tinggi
53	33.422410	77.130847	95.076602	1.233205	37.452147	rendah
54	28.968500	94.497766	50.847824	0.817559	34.505356	rendah
55	33.828114	68.471935	167.040379	3.390887	45.782613	sedang
56	21.327388	83.524744	280.846432	0.619301	52.475430	tinggi
57	22.939743	86.656687	208.808939	3.135990	52.362131	tinggi
58	20.678409	68.317314	171.018351	4.731036	49.333180	sedang
59	24.879955	85.487572	29.152948	3.089634	35.574605	rendah

	suhu	kelembaban	curah_hujan	luas_lahan	hasil_panen	kategori_hasil
60	25.830159	72.872410	184.502168	2.246765	49.573641	sedang
61	24.070235	82.130704	297.016155	3.394797	58.603726	tinggi
62	32.431063	82.173540	42.025205	2.562138	37.235465	rendah
63	25.351300	78.752114	155.498896	2.955276	45.182937	sedang
64	24.214018	63.160142	263.211922	4.736592	60.070215	tinggi
65	28.140441	89.235587	222.230585	2.237462	51.370636	tinggi
66	22.113863	71.227302	209.104722	4.825358	49.761363	sedang
67	32.032955	66.528148	210.745225	4.574078	51.939409	tinggi
68	21.118260	61.427130	107.847345	1.381060	27.919967	rendah
69	34.803304	80.681253	88.077553	0.812126	35.957739	rendah
70	31.583672	83.714753	242.808347	0.953501	50.887623	tinggi
71	22.980735	60.580574	243.034018	0.581998	44.778521	sedang
72	20.082832	77.923257	260.121696	0.924993	50.155169	tinggi
73	32.231921	67.927352	273.972166	3.573530	61.551666	tinggi
74	30.602860	82.581048	153.402720	0.820349	44.578885	sedang
75	30.935108	66.102825	150.454888	1.935390	40.263559	sedang
76	31.569055	84.182821	239.488554	4.301939	57.063185	tinggi
77	21.110670	73.535737	194.989179	0.604724	42.732552	sedang
78	25.376986	92.785550	210.590063	4.165108	52.918962	tinggi
79	21.738036	64.813233	238.737801	1.768346	50.557448	tinggi
80	32.946551	71.937322	267.001603	1.031742	55.393954	tinggi
81	29.349472	63.971573	101.398547	3.635317	38.071294	rendah
82	24.963470	92.364277	112.674886	3.330243	40.463602	sedang
83	20.953375	90.706877	28.194582	4.448624	38.851839	rendah
84	24.664735	69.027957	173.484042	3.807820	45.939976	sedang
85	24.877750	83.099442	10.782682	4.115664	35.868442	rendah
86	30.944093	88.602777	139.679405	1.769156	41.321180	sedang
87	29.563362	79.432028	162.793390	1.298478	42.432959	sedang
88	33.308191	78.537770	85.962376	3.877766	42.062269	sedang
89	27.083224	68.464830	177.249978	4.130756	47.898405	sedang

	suhu	kelembaban	curah_hujan	luas_lahan	hasil_panen	kategori_hasil
90	21.793914	63.258597	9.150075	4.957273	29.119316	rendah
91	30.698672	91.402552	11.204457	2.356780	34.569816	rendah
92	31.411776	91.514632	246.780168	2.174081	54.617398	tinggi
93	28.419158	82.158551	108.057192	3.993858	43.466135	sedang
94	31.564508	71.866043	38.118154	2.033616	31.962199	rendah
95	27.406934	72.222335	156.672978	4.688408	48.739539	sedang
96	27.840992	85.408449	230.998066	4.362857	58.682739	tinggi
97	26.413115	91.398859	64.746308	2.430473	35.289999	rendah
98	20.381287	91.048025	186.867143	3.878920	47.700317	sedang
99	21.618371	87.295644	25.604239	3.895443	36.851304	rendah

In [5]:
```
import pandas as pd

# cek jumlah missing value tiap kolom
df.isnull().sum()
```

Out[5]:
```
suhu              0
kelembaban        0
curah_hujan       0
luas_lahan        0
hasil_panen       0
kategori_hasil    0
dtype: int64
```

In [11]:
```
(df.isnull().sum() / len(df)) * 100
```

Out[11]:
```
suhu              0.0
kelembaban        0.0
curah_hujan       0.0
luas_lahan        0.0
hasil_panen       0.0
kategori_hasil    0.0
dtype: float64
```

In [12]:
```
df.isnull().sum()
```

Out[12]:
```
suhu              0
kelembaban        0
curah_hujan       0
luas_lahan        0
hasil_panen       0
kategori_hasil    0
dtype: int64
```

In [17]:
```
from sklearn.preprocessing import MinMaxScaler
```

```python
# pilih kolom numerik (tanpa kategori)
numerical_cols = ['suhu', 'kelembaban', 'curah_hujan', 'luas_lahan', 'hasil_panen']

scaler = MinMaxScaler()

# copy data biar aman
df_scaled = df.copy()

# lakukan normalisasi
df_scaled[numerical_cols] = scaler.fit_transform(df[numerical_cols])

df_scaled.head(100)
```

Out[17]:

	suhu	kelembaban	curah_hujan	luas_lahan	hasil_panen	kategori_hasil
0	0.376025	0.025010	0.646675	0.038201	0.472321	sedang
1	0.963140	0.643159	0.080283	0.529613	0.237118	rendah
2	0.740267	0.314095	0.158953	0.539120	0.251282	rendah
3	0.604399	0.512536	0.907106	0.638284	0.878241	tinggi
4	0.153354	0.920216	0.610530	0.729115	0.708320	tinggi
5	0.153330	0.247615	0.004198	0.984988	0.097737	rendah
6	0.053560	0.412212	0.097879	0.514190	0.091212	rendah
7	0.876997	0.764893	0.668472	0.316114	0.730153	tinggi
8	0.606903	0.226675	0.000000	0.799901	0.061232	rendah
9	0.715891	0.071552	0.158119	0.262715	0.210897	rendah
10	0.015348	0.288955	0.551956	0.434969	0.365084	rendah
11	0.982701	0.157627	0.697298	0.065631	0.480542	sedang
12	0.842623	0.942829	0.656756	0.011225	0.706188	tinggi
13	0.210744	0.818606	0.222548	0.971462	0.453447	sedang
14	0.179651	0.640087	0.717892	0.841693	0.793583	tinggi
15	0.181260	0.883325	0.235725	0.698261	0.395111	sedang
16	0.304393	0.814061	0.325219	0.404215	0.361950	rendah
17	0.529094	0.183527	0.752727	0.162790	0.517996	sedang
18	0.434520	0.904882	0.654392	0.145520	0.616358	sedang
19	0.291132	0.543978	0.857024	0.241621	0.680347	tinggi
20	0.617844	0.817911	0.662494	0.547922	0.783138	tinggi
21	0.136516	0.908492	0.571829	0.717338	0.640759	sedang
22	0.292065	0.317821	0.089963	0.661609	0.080341	rendah
23	0.367692	0.105344	0.368180	0.272039	0.215376	rendah
24	0.459103	0.225793	0.264104	0.963488	0.379551	rendah
25	0.794459	0.429300	0.242568	0.741210	0.331501	rendah
26	0.197838	0.828716	0.982697	0.553175	0.902242	tinggi
27	0.518372	0.872361	0.393948	0.611946	0.526575	sedang
28	0.598037	0.000000	0.900499	0.415123	0.746109	tinggi
29	0.041705	0.514760	0.635616	0.239048	0.463434	sedang

	suhu	kelembaban	curah_hujan	luas_lahan	hasil_panen	kategori_hasil
30	0.613455	0.419393	0.801783	0.349939	0.689033	tinggi
31	0.168135	0.219839	0.505157	0.761647	0.485102	sedang
32	0.060660	0.115371	0.580555	0.000000	0.225324	rendah
33	0.961277	0.337860	0.494883	0.104168	0.339634	rendah
34	0.978341	0.956329	0.193079	0.032383	0.228433	rendah
35	0.818121	0.323134	0.728321	0.026980	0.566592	sedang
36	0.304771	0.522979	0.279912	0.861651	0.400384	sedang
37	0.093900	0.711217	0.019548	0.706133	0.216721	rendah
38	0.691599	0.364441	0.650168	0.471033	0.666255	tinggi
39	0.442884	0.985830	0.174671	0.085483	0.196067	rendah
40	0.118729	0.976292	0.949649	0.488901	0.923191	tinggi
41	0.498953	0.250159	0.963324	0.470313	0.778462	tinggi
42	0.029415	0.500968	0.923665	0.162695	0.713281	tinggi
43	0.920961	0.300324	0.370660	0.429724	0.368214	rendah
44	0.258067	0.283937	0.010553	0.393512	0.000000	rendah
45	0.669476	0.030586	0.937324	0.616176	0.781704	tinggi
46	0.312003	0.615728	0.429569	0.635891	0.501926	sedang
47	0.524317	0.506517	0.976244	0.031667	0.878385	tinggi
48	0.551465	0.045496	0.973163	0.369035	0.778990	tinggi
49	0.182738	0.277608	0.860868	0.626431	0.776848	tinggi
50	0.982369	0.920931	0.293797	0.500704	0.507496	sedang
51	0.784225	0.237673	0.385827	0.862705	0.454494	sedang
52	0.951712	0.140945	0.858966	0.660068	0.844395	tinggi
53	0.906192	0.493002	0.316612	0.152176	0.305441	rendah
54	0.603627	1.000000	0.166937	0.057550	0.220513	rendah
55	0.933753	0.240220	0.560146	0.643395	0.545529	sedang
56	0.084546	0.679661	0.945280	0.012414	0.738419	tinggi
57	0.194077	0.771093	0.701496	0.585366	0.735154	tinggi
58	0.040459	0.235706	0.573608	0.948495	0.647858	sedang
59	0.325881	0.736963	0.093518	0.574812	0.251330	rendah

	suhu	kelembaban	curah_hujan	luas_lahan	hasil_panen	kategori_hasil
60	0.390431	0.368685	0.619239	0.382924	0.654788	sedang
61	0.270875	0.638965	1.000000	0.644286	0.915039	tinggi
62	0.838847	0.640215	0.137080	0.454722	0.299196	rendah
63	0.357901	0.540333	0.521088	0.544224	0.528246	sedang
64	0.280642	0.085152	0.885602	0.949760	0.957304	tinggi
65	0.547374	0.846380	0.746917	0.380806	0.706578	tinggi
66	0.137973	0.320658	0.702497	0.969968	0.660198	sedang
67	0.811803	0.183475	0.708049	0.912762	0.722971	tinggi
68	0.070339	0.034559	0.359830	0.185837	0.030720	rendah
69	1.000000	0.596650	0.292926	0.056313	0.262372	rendah
70	0.781282	0.685208	0.816554	0.088499	0.692658	tinggi
71	0.196862	0.009845	0.817318	0.003922	0.516591	sedang
72	0.000000	0.516135	0.875145	0.082009	0.671548	tinggi
73	0.825319	0.224322	0.922016	0.684976	1.000000	tinggi
74	0.714653	0.652112	0.513995	0.058185	0.510837	sedang
75	0.737223	0.171058	0.504019	0.312036	0.386467	sedang
76	0.780289	0.698873	0.805320	0.850806	0.870640	tinggi
77	0.069824	0.388049	0.654728	0.009096	0.457625	sedang
78	0.359646	0.950015	0.707524	0.819655	0.751202	tinggi
79	0.112442	0.133411	0.802779	0.274007	0.683142	tinggi
80	0.873866	0.341386	0.898427	0.106311	0.822532	tinggi
81	0.629507	0.108840	0.338006	0.699043	0.323285	rendah
82	0.331554	0.937716	0.376167	0.629589	0.392233	sedang
83	0.059138	0.889331	0.090275	0.884201	0.345781	rendah
84	0.311261	0.256452	0.581952	0.738315	0.550064	sedang
85	0.325731	0.667245	0.031351	0.808399	0.259798	rendah
86	0.737834	0.827906	0.467553	0.274191	0.416948	sedang
87	0.644037	0.560181	0.545774	0.167036	0.448990	sedang
88	0.898433	0.534075	0.285768	0.754239	0.438307	sedang
89	0.475555	0.240013	0.594697	0.811835	0.606507	sedang

	suhu	kelembaban	curah_hujan	luas_lahan	hasil_panen	kategori_hasil
90	0.116238	0.088026	0.025826	1.000000	0.065285	rendah
91	0.721162	0.909640	0.032779	0.407970	0.222371	rendah
92	0.769605	0.912912	0.829995	0.366377	0.800151	tinggi
93	0.566308	0.639778	0.360540	0.780668	0.478767	sedang
94	0.779980	0.339305	0.123858	0.334398	0.147218	rendah
95	0.497545	0.349707	0.525062	0.938790	0.630749	sedang
96	0.527032	0.734653	0.776587	0.864675	0.917316	tinggi
97	0.430033	0.909533	0.213971	0.424747	0.243127	rendah
98	0.020275	0.899291	0.627242	0.754501	0.600798	sedang
99	0.104313	0.789746	0.081509	0.758263	0.288125	rendah

In [16]:
```
# install dulu packagenya
!pip install scikit-learn
```

```
Requirement already satisfied: scikit-learn in c:\users\asus\appdata\local\programs
\python\python313\lib\site-packages (1.8.0)
Requirement already satisfied: numpy>=1.24.1 in c:\users\asus\appdata\local\programs
\python\python313\lib\site-packages (from scikit-learn) (2.4.1)
Requirement already satisfied: scipy>=1.10.0 in c:\users\asus\appdata\local\programs
\python\python313\lib\site-packages (from scikit-learn) (1.17.1)
Requirement already satisfied: joblib>=1.3.0 in c:\users\asus\appdata\local\programs
\python\python313\lib\site-packages (from scikit-learn) (1.5.3)
Requirement already satisfied: threadpoolctl>=3.2.0 in c:\users\asus\appdata\local\p
rograms\python\python313\lib\site-packages (from scikit-learn) (3.6.0)
[notice] A new release of pip is available: 25.3 -> 26.0.1
[notice] To update, run: python.exe -m pip install --upgrade pip
```

In [19]:
```
#untuk menyamakan skala nya
df_scaled[numerical_cols].describe()
```

Out[19]:

	suhu	kelembaban	curah_hujan	luas_lahan	hasil_panen
count	100.000000	100.000000	100.000000	100.000000	100.000000
mean	0.473482	0.501564	0.520349	0.488423	0.512198
std	0.303138	0.299491	0.297897	0.300634	0.247695
min	0.000000	0.000000	0.000000	0.000000	0.000000
25%	0.191242	0.240168	0.275960	0.240978	0.318824
50%	0.467329	0.509526	0.565988	0.507447	0.509167
75%	0.738442	0.775756	0.758692	0.739038	0.709560
max	1.000000	1.000000	1.000000	1.000000	1.000000

In [20]:

```
print("=== Sebelum Normalisasi ===")
print(df[numerical_cols].describe())

print("\n=== Setelah Normalisasi ===")
print(df_scaled[numerical_cols].describe())
```

```
=== Sebelum Normalisasi ===
             suhu  kelembaban  curah_hujan  luas_lahan  hasil_panen
count  100.000000  100.000000   100.000000  100.000000   100.000000
mean    27.052711   77.424110   155.280399    2.710170    44.626107
std      4.462341   10.258894    88.027874    1.320535     8.594415
min     20.082832   60.243325     1.518475    0.564771    26.854072
25%     22.898011   68.470158    83.063959    1.623267    37.916507
50%     26.962137   77.696870   168.766480    2.793732    44.520929
75%     30.953047   86.816426   225.710083    3.810999    51.474109
max     34.803304   94.497766   297.016155    4.957273    61.551666

=== Setelah Normalisasi ===
             suhu  kelembaban  curah_hujan  luas_lahan  hasil_panen
count  100.000000  100.000000   100.000000  100.000000   100.000000
mean     0.473482    0.501564     0.520349    0.488423     0.512198
std      0.303138    0.299491     0.297897    0.300634     0.247695
min      0.000000    0.000000     0.000000    0.000000     0.000000
25%      0.191242    0.240168     0.275960    0.240978     0.318824
50%      0.467329    0.509526     0.565988    0.507447     0.509167
75%      0.738442    0.775756     0.758692    0.739038     0.709560
max      1.000000    1.000000     1.000000    1.000000     1.000000
```

In [21]:

```
print("=== Sebelum ===")
print(df[numerical_cols].head())

print("\n=== Sesudah ===")
print(df_scaled[numerical_cols].head())
```

```
=== Sebelum ===
        suhu  kelembaban  curah_hujan  luas_lahan  hasil_panen
0  25.618102   61.100021   192.609494    0.732568    43.242477
1  34.260715   82.274364    25.241989    2.891096    35.081486
2  30.979909   71.002459    48.488614    2.932858    35.572948
3  28.979877   77.799974   269.566257    3.368435    57.326919
4  22.340280   91.764827   181.928718    3.767411    51.431076

=== Sesudah ===
       suhu  kelembaban  curah_hujan  luas_lahan  hasil_panen
0  0.376025    0.025010     0.646675    0.038201     0.472321
1  0.963140    0.643159     0.080283    0.529613     0.237118
2  0.740267    0.314095     0.158953    0.539120     0.251282
3  0.604399    0.512536     0.907106    0.638284     0.878241
4  0.153354    0.920216     0.610530    0.729115     0.708320
```

In [22]:
```
from sklearn.model_selection import train_test_split

X = df_scaled.drop('kategori_hasil', axis=1)
y = df_scaled['kategori_hasil']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

In [23]:
```
from sklearn.preprocessing import LabelEncoder

le = LabelEncoder()
y = le.fit_transform(df_scaled['kategori_hasil'])
```

In [24]:
```
X = df_scaled.drop('kategori_hasil', axis=1)

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

In [25]:
```
from sklearn.tree import DecisionTreeClassifier

model = DecisionTreeClassifier(random_state=42)
model.fit(X_train, y_train)
```

Out[25]:
```
▾ DecisionTreeClassifier
▸ Parameters
```

In [26]:
```
y_pred = model.predict(X_test)
```

In [27]:
```
from sklearn.metrics import accuracy_score

accuracy = accuracy_score(y_test, y_pred)
print("Akurasi:", accuracy)
```

```
Akurasi: 1.0
```

In [28]:
```
from sklearn.metrics import confusion_matrix

cm = confusion_matrix(y_test, y_pred)
print("Confusion Matrix:\n", cm)
```

```
Confusion Matrix:
 [[8 0 0]
 [0 4 0]
 [0 0 8]]
```

In [29]:
```
model = DecisionTreeClassifier(max_depth=3, random_state=42)
```

Performa Model

Data dibagi menjadi data training dan testing dengan rasio 80:20. Model klasifikasi yang digunakan adalah Decision Tree.

Berdasarkan hasil evaluasi, model menghasilkan akurasi sebesar 1.0 (100%), yang menunjukkan bahwa seluruh data uji berhasil diprediksi dengan benar.

Confusion matrix juga menunjukkan tidak adanya kesalahan klasifikasi pada setiap kelas. Hal ini menandakan bahwa model memiliki performa yang sangat baik.

Namun, hasil yang terlalu sempurna ini dapat mengindikasikan kemungkinan overfitting, sehingga perlu dilakukan evaluasi lebih lanjut untuk memastikan kemampuan generalisasi model terhadap data baru.

In [33]:
```
#lanjut ke bagian 4
```

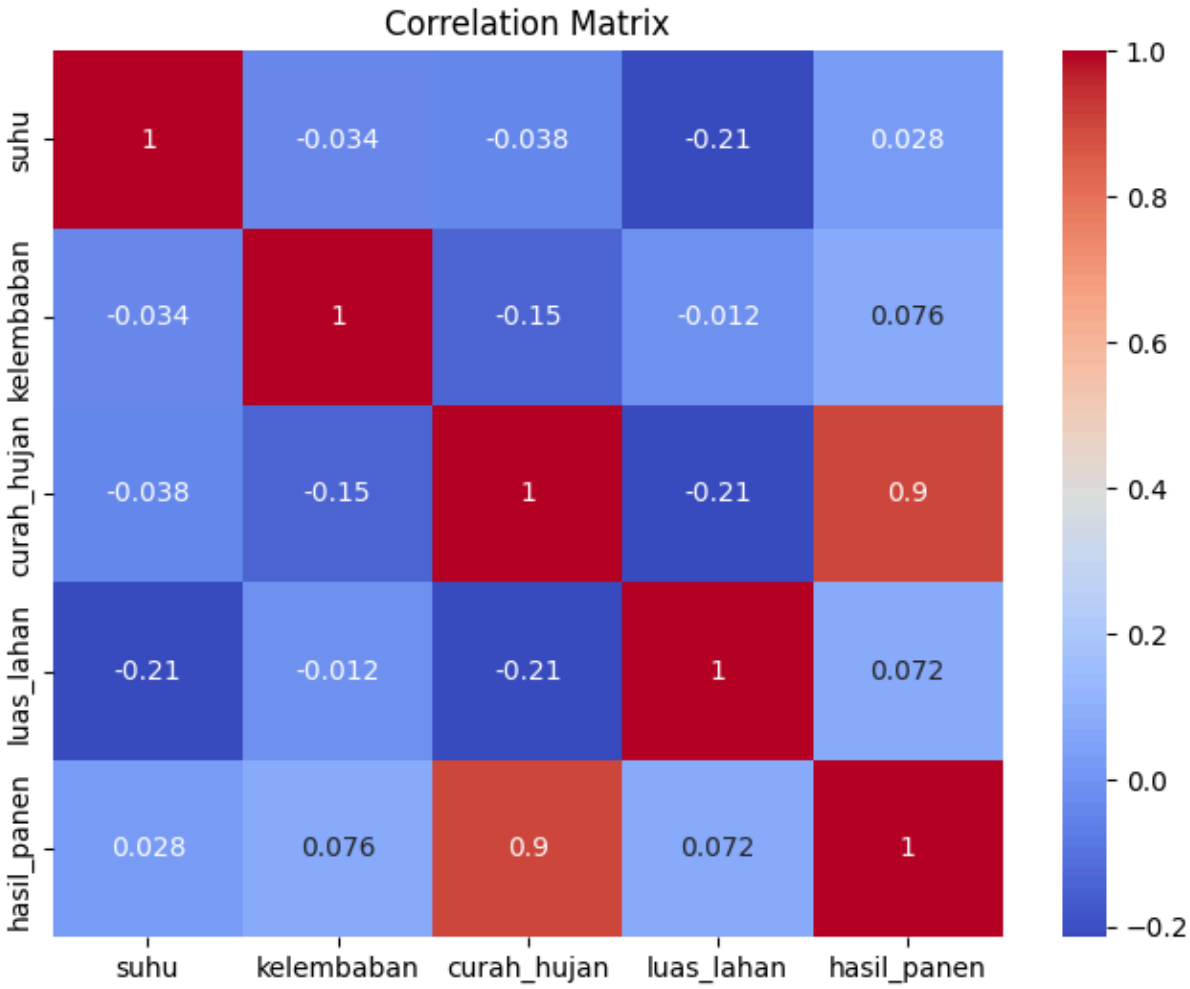
In [34]:
```
df.describe()
```

Out[34]:

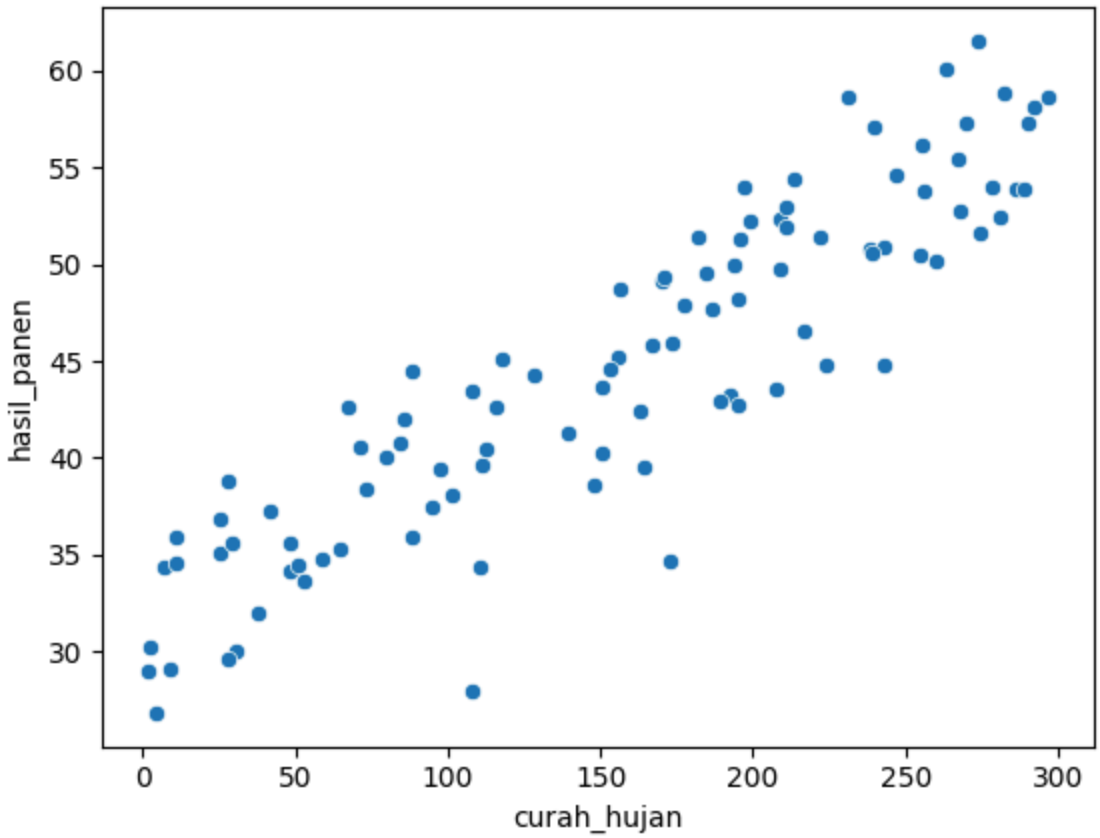
	suhu	kelembaban	curah_hujan	luas_lahan	hasil_panen
count	100.000000	100.000000	100.000000	100.000000	100.000000
mean	27.052711	77.424110	155.280399	2.710170	44.626107
std	4.462341	10.258894	88.027874	1.320535	8.594415
min	20.082832	60.243325	1.518475	0.564771	26.854072
25%	22.898011	68.470158	83.063959	1.623267	37.916507
50%	26.962137	77.696870	168.766480	2.793732	44.520929
75%	30.953047	86.816426	225.710083	3.810999	51.474109
max	34.803304	94.497766	297.016155	4.957273	61.551666

In [35]:
```
import seaborn as sns
import matplotlib.pyplot as plt

plt.figure(figsize=(8,6))
```

```
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm')
plt.title("Correlation Matrix")
plt.show()
```



In [36]:
```
sns.scatterplot(x='curah_hujan', y='hasil_panen', data=df)
plt.show()
```



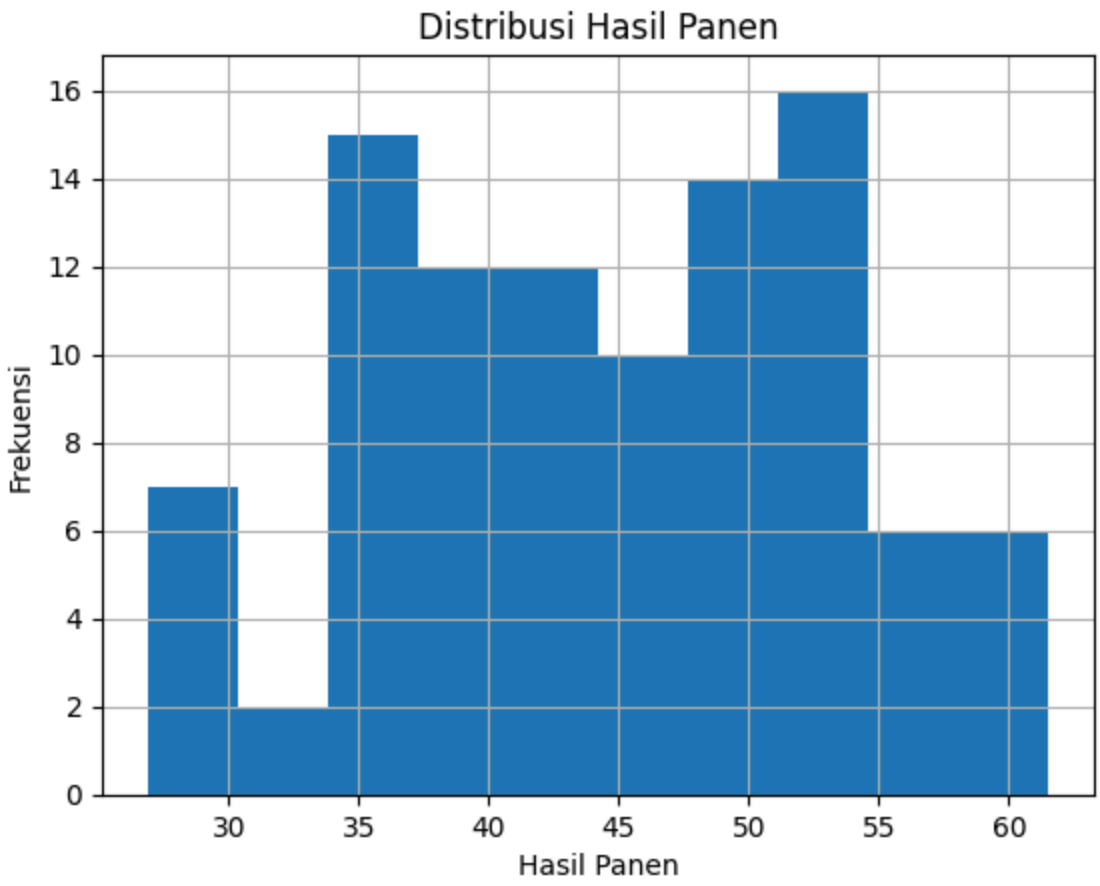
Insight Data

1. Curah hujan memiliki korelasi positif yang sangat kuat terhadap hasil panen (r ≈ 0.90). Hal ini menunjukkan bahwa curah hujan merupakan faktor utama yang mempengaruhi peningkatan hasil panen.

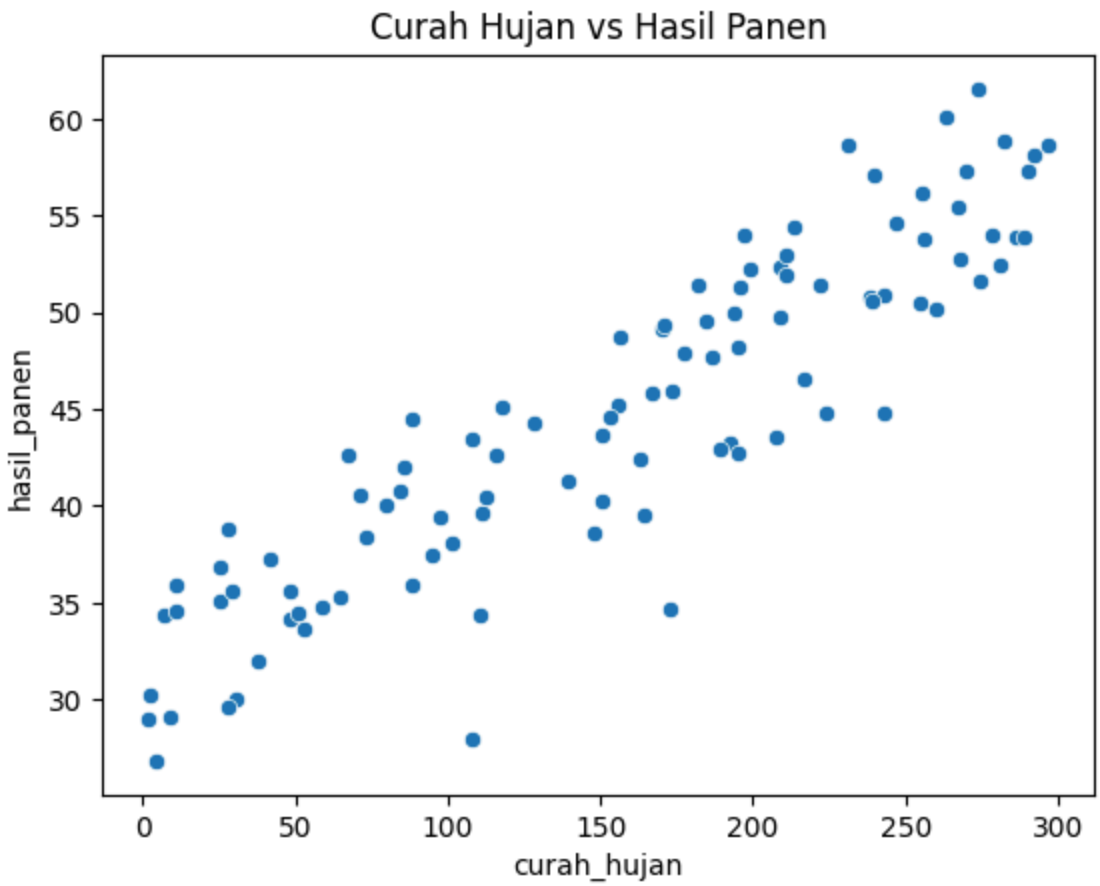
2. Luas lahan tidak memiliki hubungan yang signifikan terhadap hasil panen (r ≈ 0.07). Hal ini terlihat dari nilai korelasi yang sangat rendah serta distribusi luas lahan yang relatif seragam pada setiap kategori hasil panen.

3. Suhu dan kelembaban juga tidak menunjukkan pengaruh yang signifikan terhadap hasil panen (r ≈ 0.03 dan r ≈ 0.08). Dengan demikian, variabel-variabel tersebut memiliki kontribusi yang relatif kecil dibandingkan curah hujan.

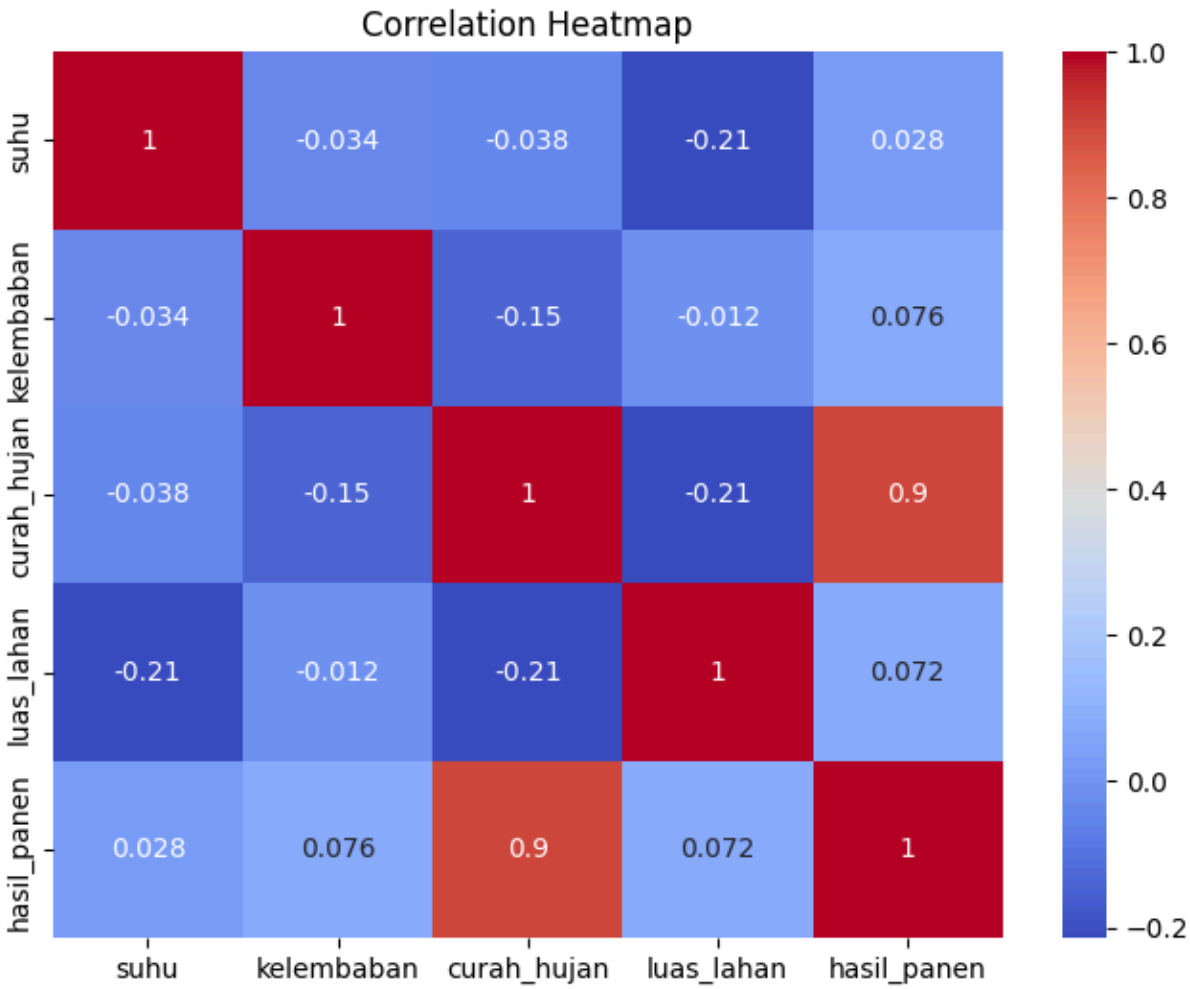
In [40]:
```
# Histogram digunakan untuk melihat distribusi data hasil panen, sehingga dapat dik
import matplotlib.pyplot as plt

df['hasil_panen'].hist()
plt.title("Distribusi Hasil Panen")
plt.xlabel("Hasil Panen")
plt.ylabel("Frekuensi")
plt.show()
```



In [41]:
```
#Scatter plot digunakan untuk melihat hubungan antara curah hujan dan hasil panen.
import seaborn as sns

sns.scatterplot(x='curah_hujan', y='hasil_panen', data=df)
plt.title("Curah Hujan vs Hasil Panen")
plt.show()
```



In [42]:
```
#Heatmap digunakan untuk menampilkan korelasi antar variabel numerik secara visual,
plt.figure(figsize=(8,6))
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()
```



Alasan Pemilihan Visualisasi

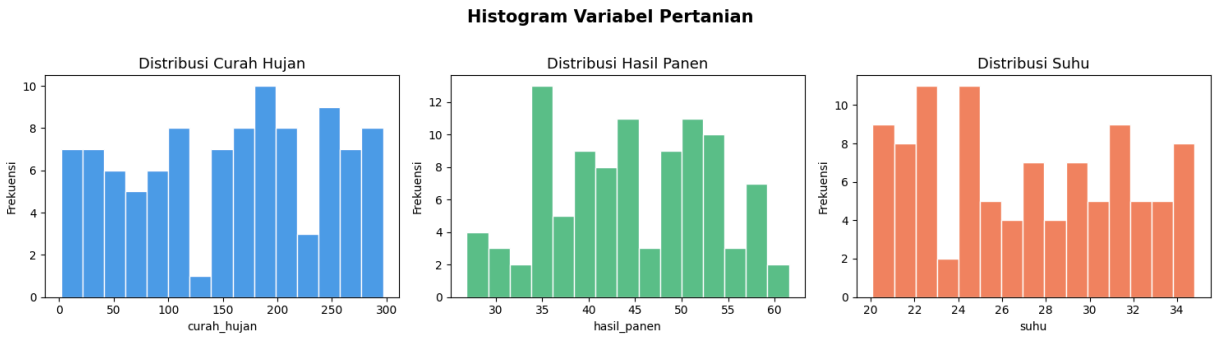
Histogram digunakan untuk memahami distribusi data. Scatter plot digunakan untuk melihat hubungan antar variabel, khususnya antara curah hujan dan hasil panen. Heatmap digunakan untuk menganalisis korelasi antar seluruh variabel numerik secara menyeluruh.

Interpretasi Hasil

Distribusi hasil panen pada histogram menunjukkan bahwa data tersebar secara cukup merata tanpa adanya outlier yang ekstrem. Distribusi hasil panen tidak mengikuti pola normal, melainkan menunjukkan kecenderungan memiliki lebih dari satu puncak, khususnya pada rentang nilai sekitar 35–40 dan 50–55. Pola ini kemungkinan berkaitan dengan adanya kategori hasil panen (rendah, sedang, dan tinggi), di mana data cenderung terkumpul pada beberapa kelompok tertentu.

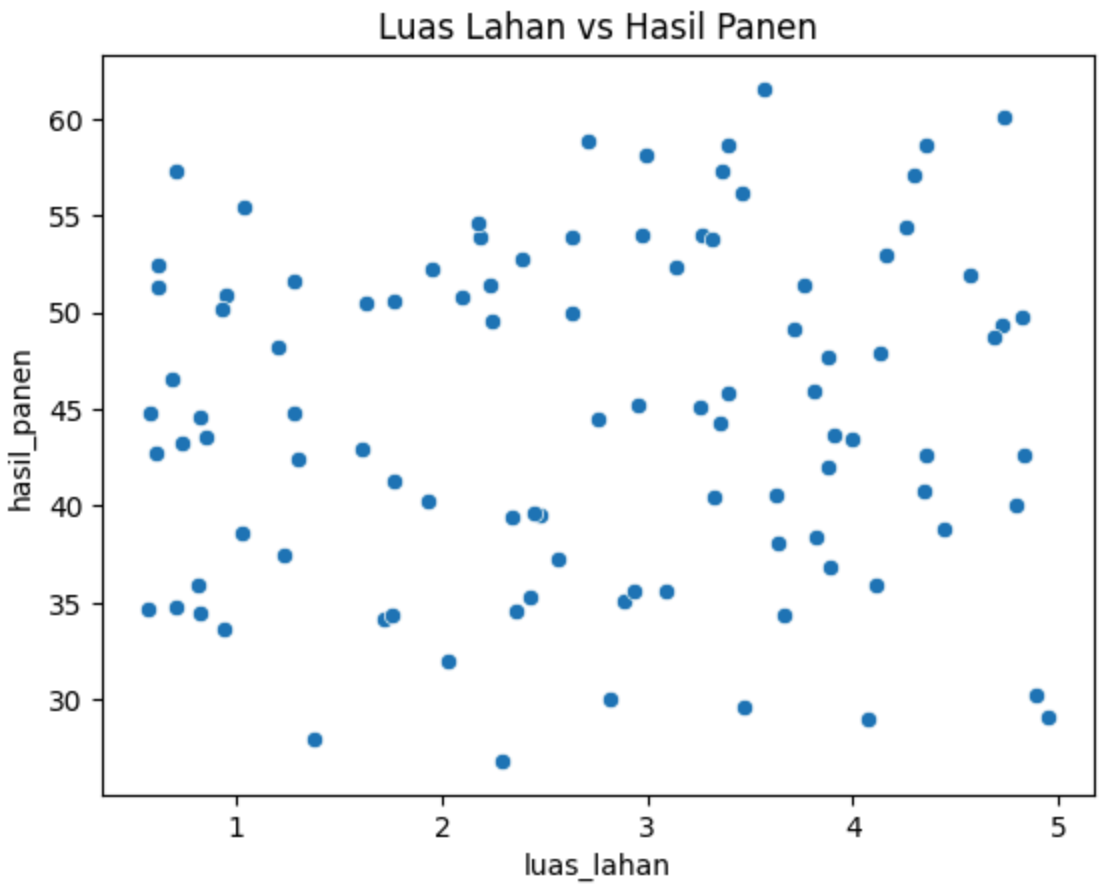
In [43]:
```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv('data_pertanian.csv')

fig, axes = plt.subplots(1, 3, figsize=(15, 4))
```

```
variables = ['curah_hujan', 'hasil_panen', 'suhu']
titles = ['Distribusi Curah Hujan', 'Distribusi Hasil Panen', 'Distribusi Suhu']
colors = ['#4C9BE8', '#5DBF8A', '#F4845F']

for i, (var, title, color) in enumerate(zip(variables, titles, colors)):
    axes[i].hist(df[var], bins=15, color=color, edgecolor='white')
    axes[i].set_title(title, fontsize=13)
    axes[i].set_xlabel(var)
    axes[i].set_ylabel('Frekuensi')

plt.suptitle('Histogram Variabel Pertanian', fontsize=15, fontweight='bold', y=1.02
plt.tight_layout()
plt.savefig('histogram.png', dpi=150, bbox_inches='tight')
plt.show()
```

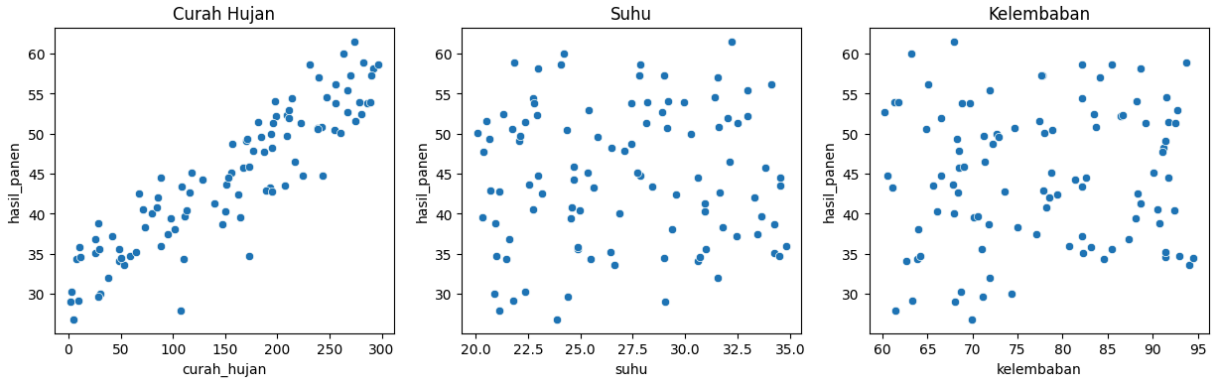


In [44]:
```
# permasalahan dan analisis tambahan
```

In [46]:
```
#Kenapa luas lahan tidak berpengaruh?
sns.scatterplot(x='luas_lahan', y='hasil_panen', data=df)
plt.title("Luas Lahan vs Hasil Panen")
plt.show()
```

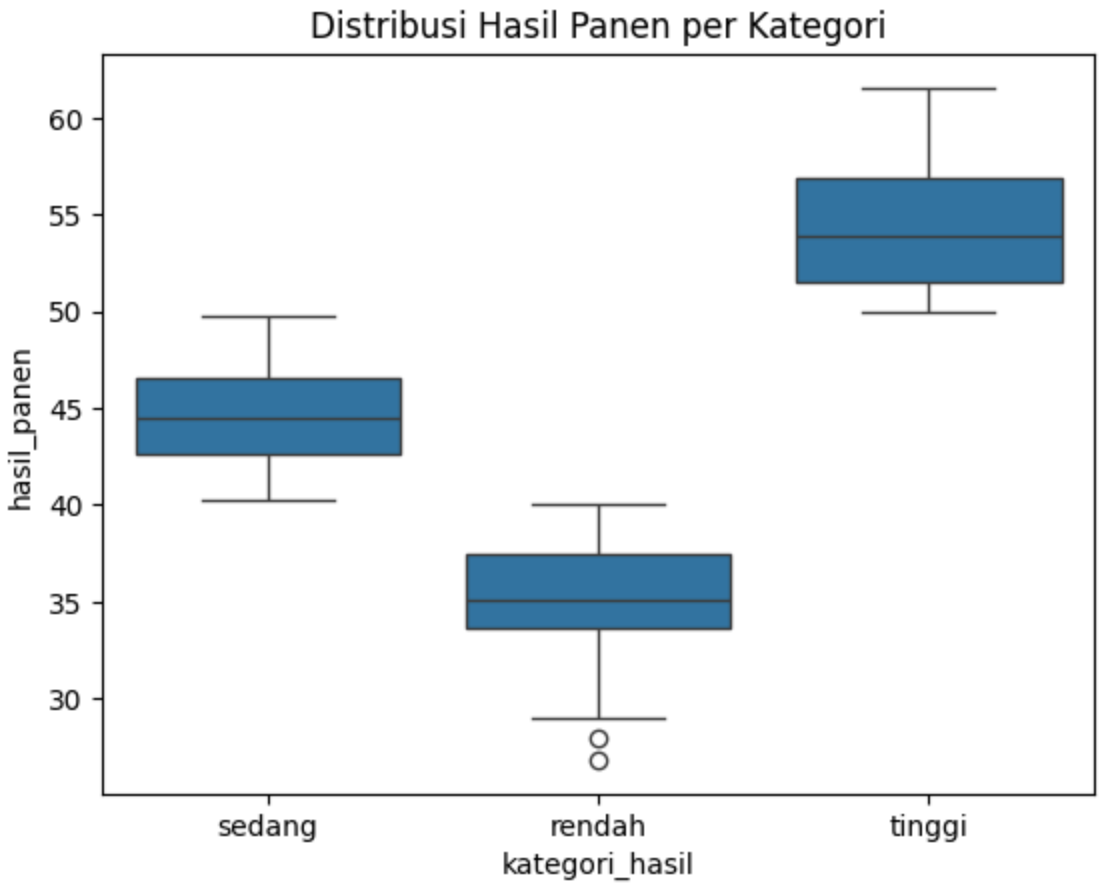


In [47]:
```
# Seberapa kuat dominasi curah hujan?
fig, axes = plt.subplots(1, 3, figsize=(15,4))

sns.scatterplot(x='curah_hujan', y='hasil_panen', data=df, ax=axes[0])
axes[0].set_title("Curah Hujan")

sns.scatterplot(x='suhu', y='hasil_panen', data=df, ax=axes[1])
axes[1].set_title("Suhu")

sns.scatterplot(x='kelembaban', y='hasil_panen', data=df, ax=axes[2])
axes[2].set_title("Kelembaban")

plt.show()
```

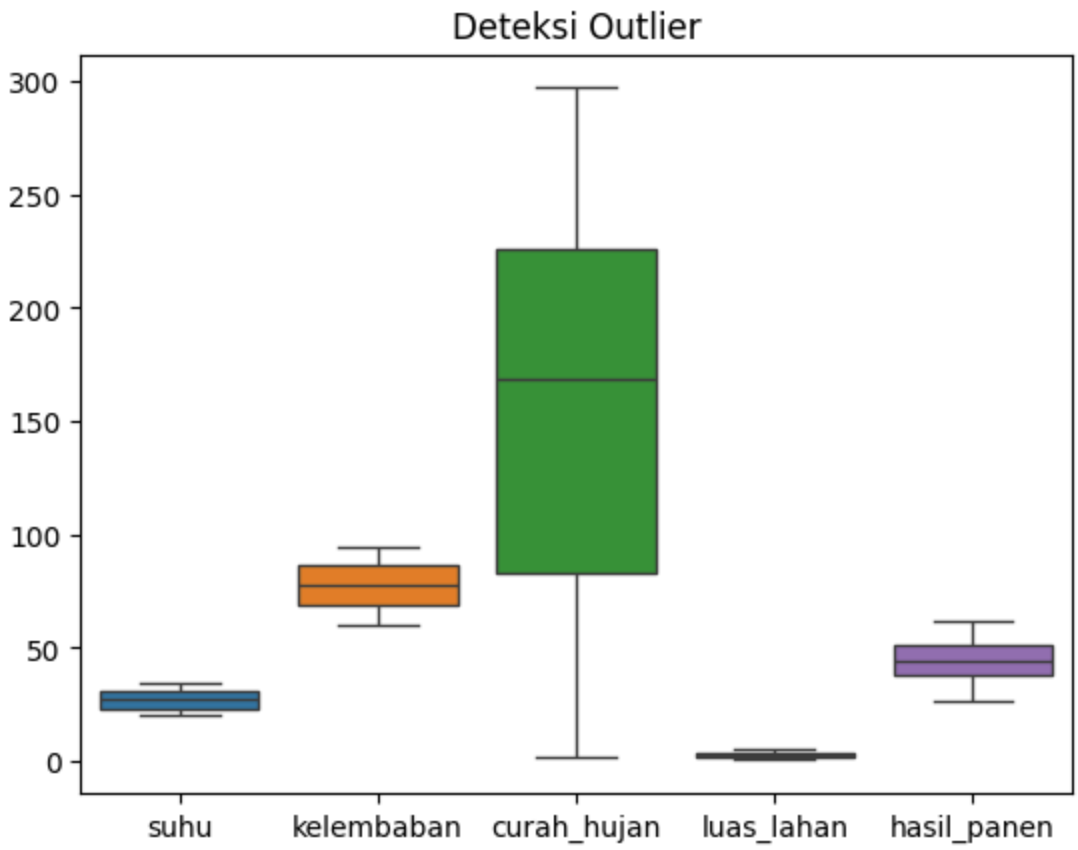


In [48]:
```
# Apakah kategori_hasil benar-benar terpisah?
sns.boxplot(x='kategori_hasil', y='hasil_panen', data=df)
```

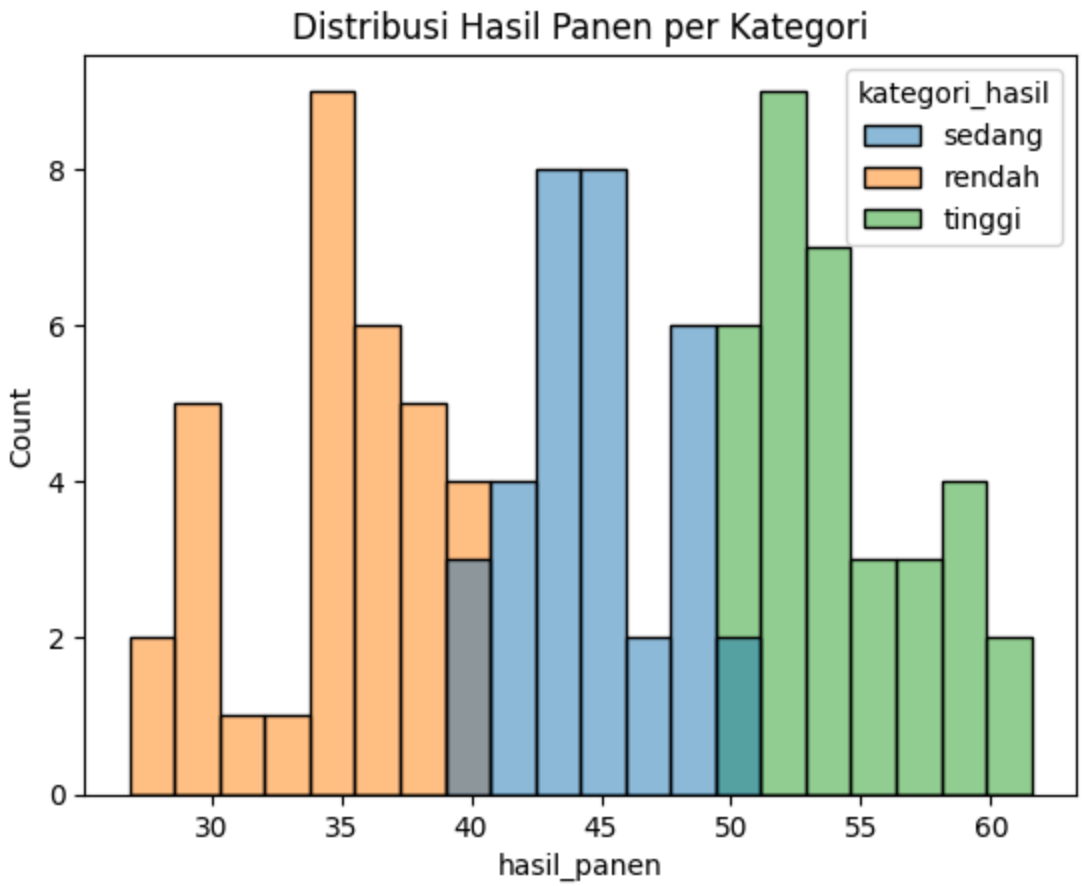
```
plt.title("Distribusi Hasil Panen per Kategori")
plt.show()
```



In [50]:
```
# Apakah ada outlier?
sns.boxplot(data=df[['suhu','kelembaban','curah_hujan','luas_lahan','hasil_panen']]
plt.title("Deteksi Outlier")
plt.show()
```



In [51]:
```
# Apakah distribusi tiap kategori beda?
sns.histplot(data=df, x='hasil_panen', hue='kategori_hasil', bins=20)
plt.title("Distribusi Hasil Panen per Kategori")
plt.show()
```



In [1]:
```
# Visualisasi tambahan menggunakan Tableau
```

In [ ]: